



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Personal Statement

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May. 29th, 2026

Southern University of Science and Technology



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01

PART ONE

Basic Information



南方科技大学
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Basic Information



Name: Yudong Li

University: Southern University of Science and Technology

Major: Intelligent Manufacturing and Robotics

Undergraduate GPA: 4.03/5.00

Master's GPA: 3.48/4.00

Undergraduate Comprehensive ranking	Master's Comprehensive ranking	CET-4	CET-6
4/107	Top 30%	558	467

02

PART TWO

Internship Experience



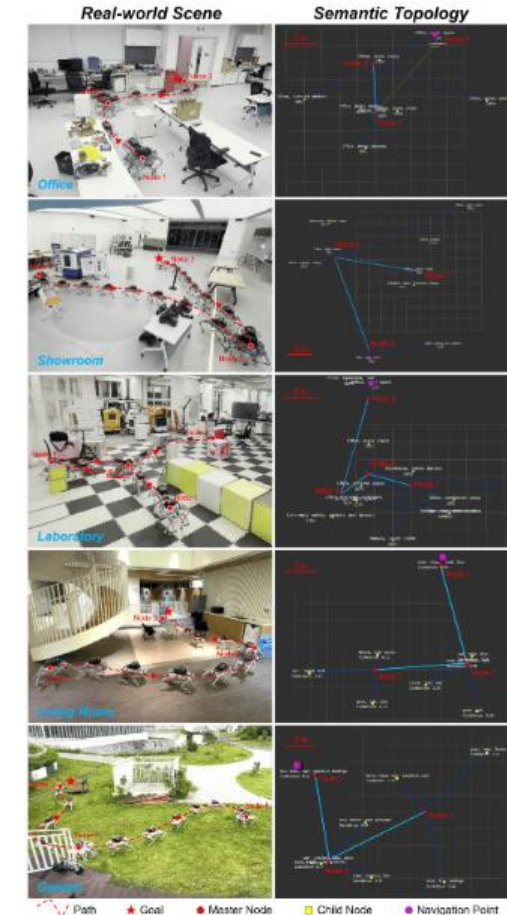
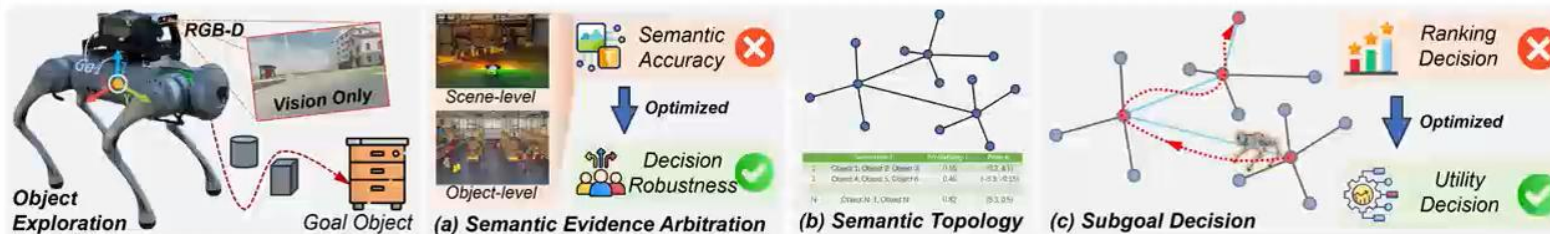
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Research Intern in The Hong Kong University of Science and Technology (Guangzhou)

From Language Instructions to Real-World Execution: Open-Vocabulary Semantic Exploration for Legged Robots

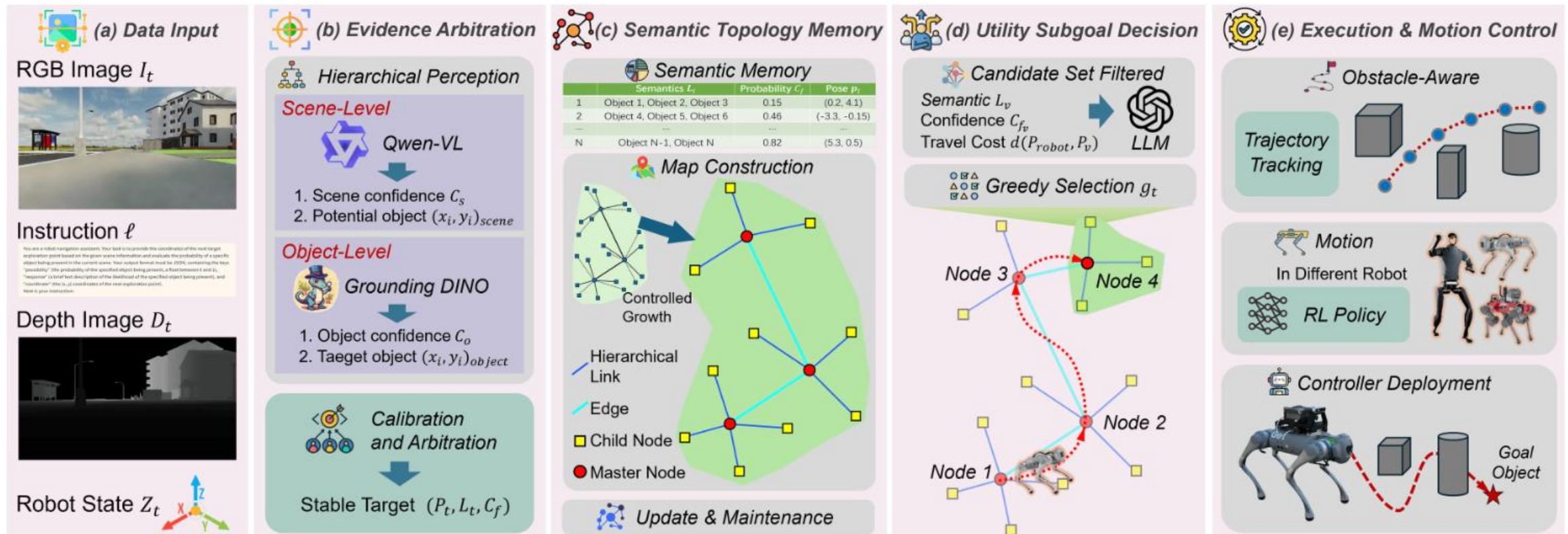
Decision-Driven Semantic Object Exploration for Legged Robots via Confidence-Calibrated Perception and Topological Subgoal Selection

Paper Number: 2247



Research Intern in The Hong Kong University of Science and Technology (Guangzhou)

From Language Instructions to Real-World Execution: Open-Vocabulary Semantic Exploration for Legged Robots



<https://arxiv.org/pdf/2509.20739>

Perception and Planning Algorithm Intern, AgiBot(智元机器人)



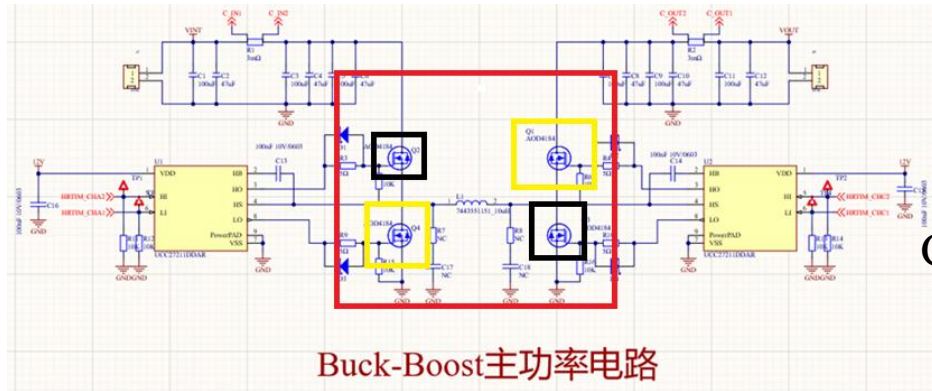
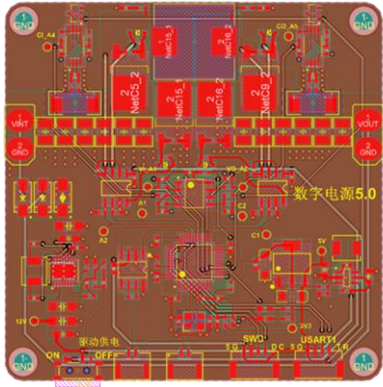
Demo footage recorded by the author at the AgiBot 2026 Partner Conference.

The robot is still under development, and no official public video is currently available.

This clip is used only to illustrate the platform and application context.

Embedded Software Intern, DJI

Digital power supply based on supercapacitors



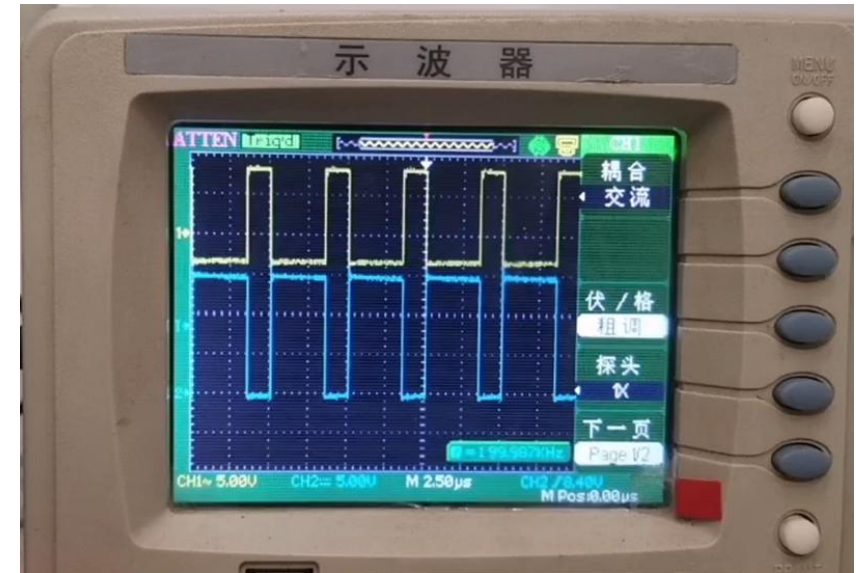
Main circuit schematic of the digital supply

$$V_{out} = \frac{D_{Buck}}{1 - D_{Boost}} V_{in}$$

$$D_{Buck} = D_{Boost}$$

$$V_{out} = \frac{D}{1 - D} V_{in}$$

Control Principle of the digital supply



Waveform of PWM output

03

PART THREE

Research Experience

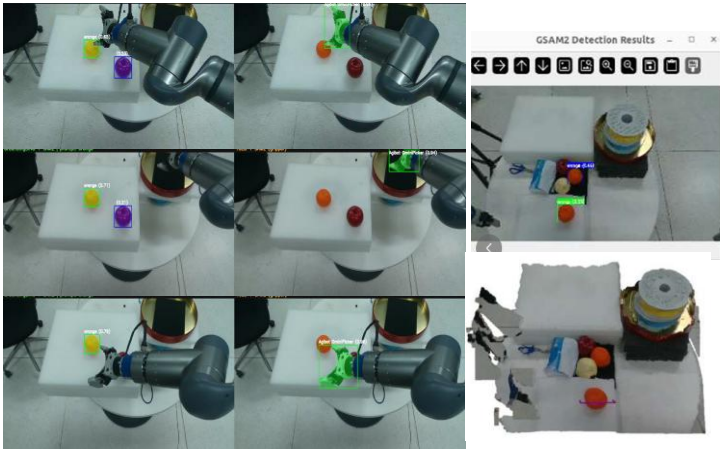


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Research on Cross-Configuration Manipulator Trajectory Planning Based on GNN-Diffusion Model

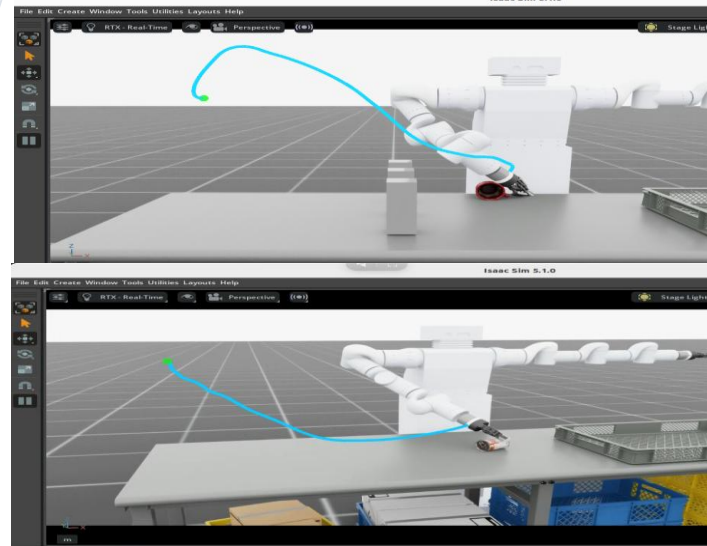
Research Team Member in Southern University of Science and Technology, Intelligent Robotics and Control Lab | Sep 2024 - Present

Perception



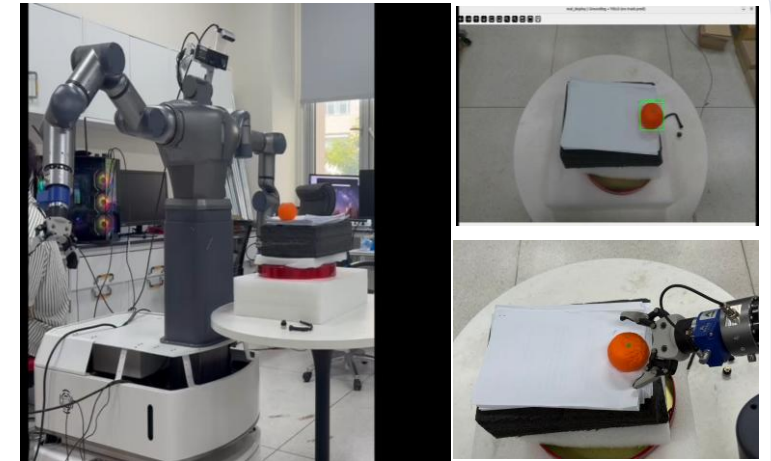
GroundingDino+SAM2+Yolo and grasp perception detect targets and estimate grasp poses by AnyGrasp from RGB-D observations.

IsaacSim Simulation Validation



GNN-conditioned diffusion generates smooth joint-space trajectories and validates cross-configuration execution in Isaac Sim.

Real-World Experiments



The planned trajectory is deployed on the real manipulator;
MPPI-MPC recovery improves robustness under execution deviations and target changes.



- URDF-based GNN morphology encoding supports cross-configuration generalization.

- B-spline diffusion planning improves trajectory smoothness and feasibility.

- Closed-loop recovery enables robust sim-to-real grasping execution.

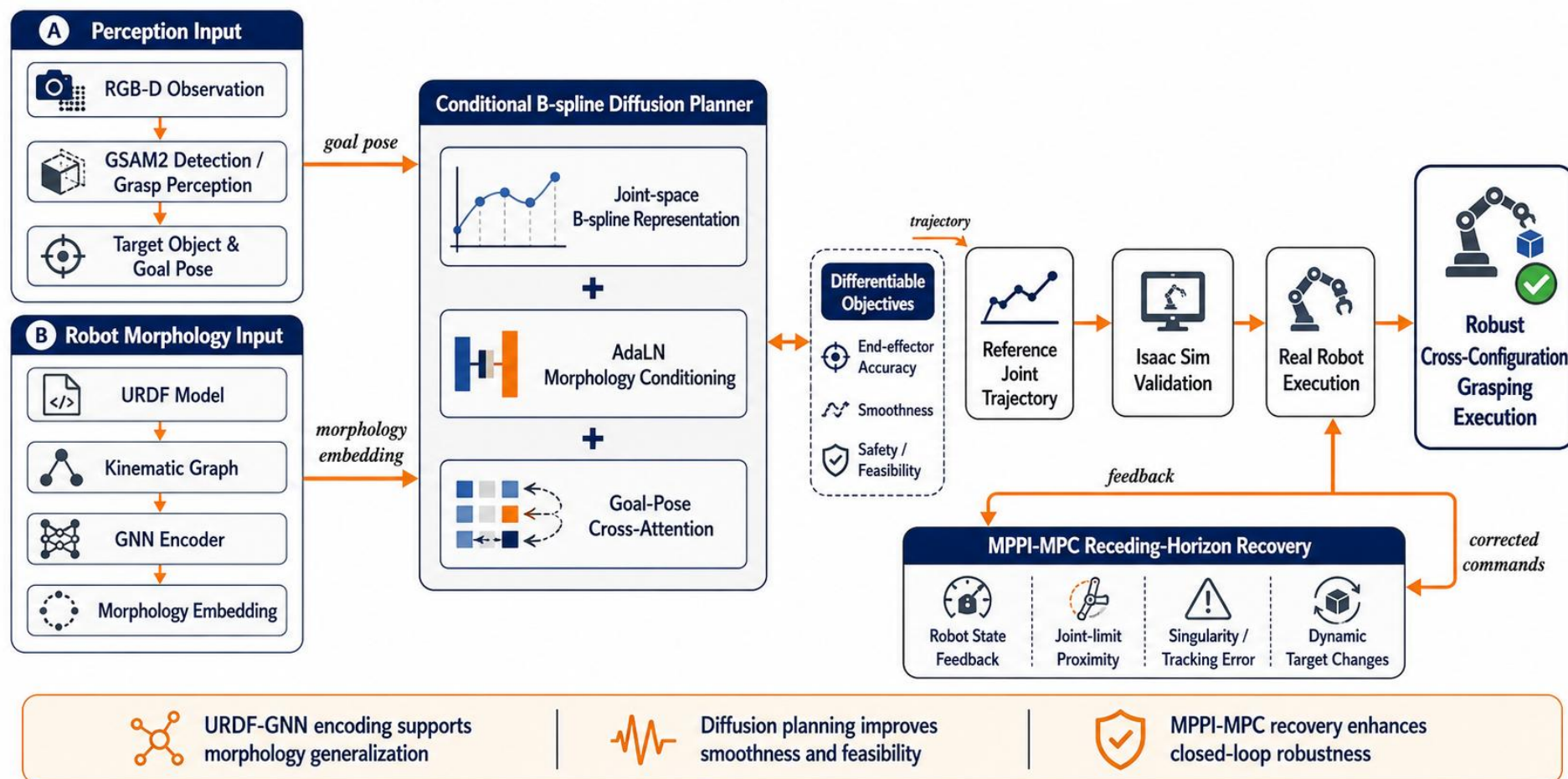
Research Experience

Research on Cross-Configuration Manipulator Trajectory Planning Based on GNN-Diffusion Models

Research Team Member in Southern University of Science and Technology, Intelligent Robotics and Control Lab | Sep 2024 - Present

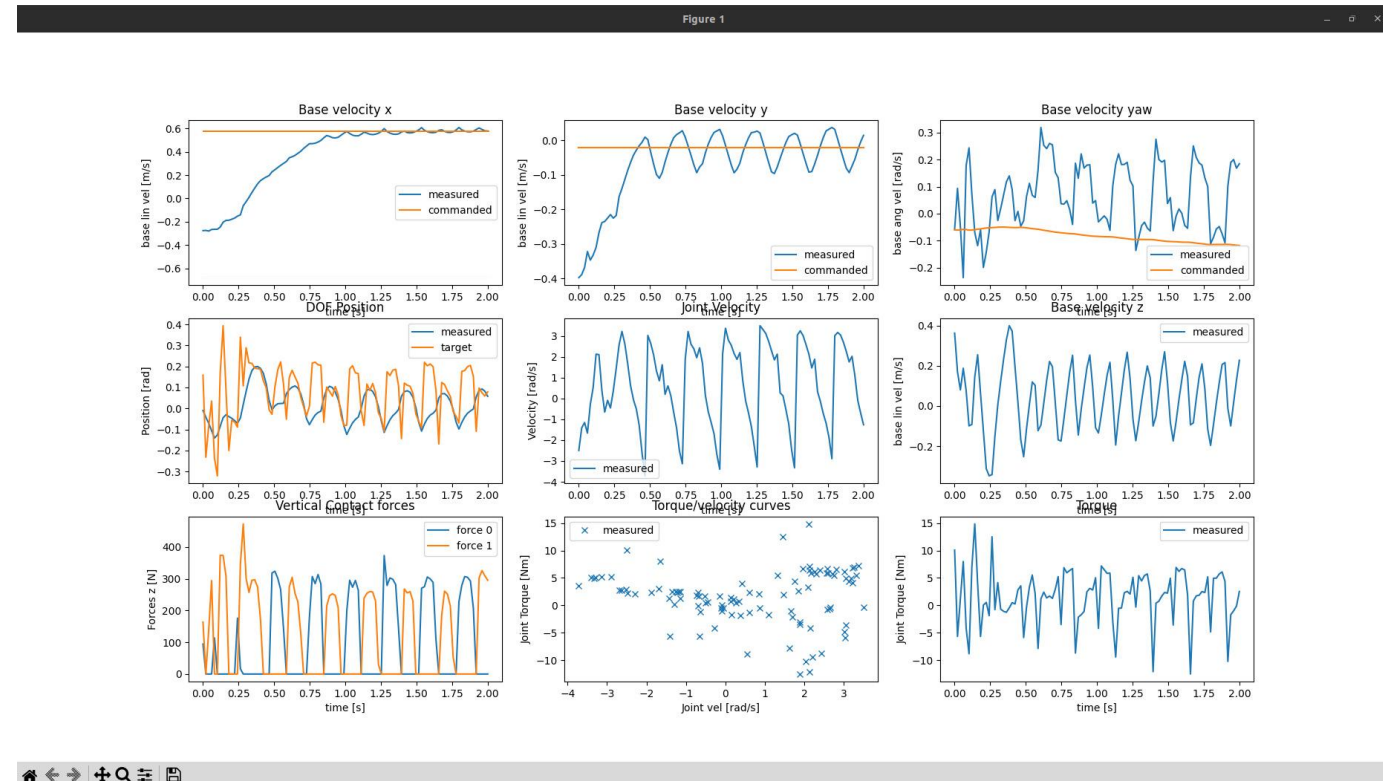
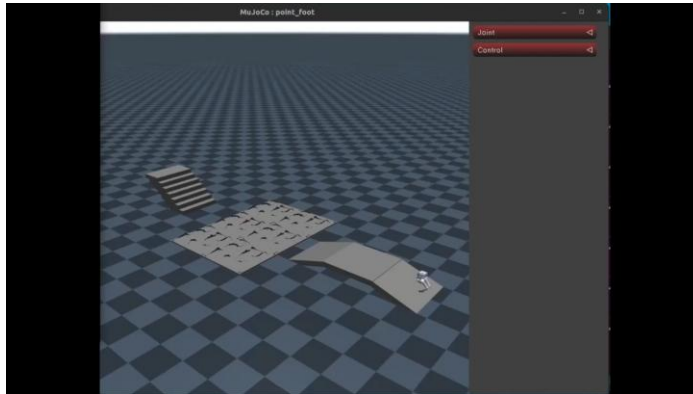
Program Framework

Cross-Configuration Manipulator Trajectory Planning Based on GNN-Diffusion Models

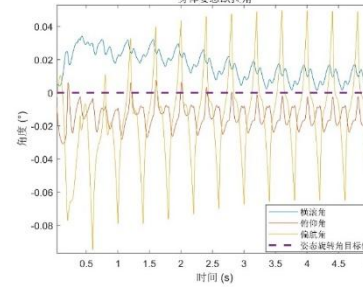
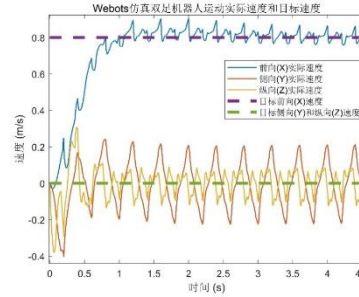
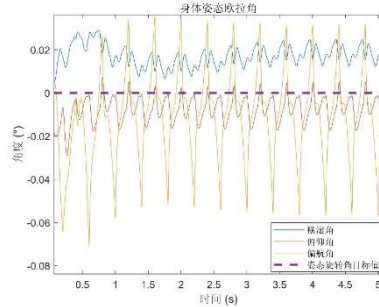
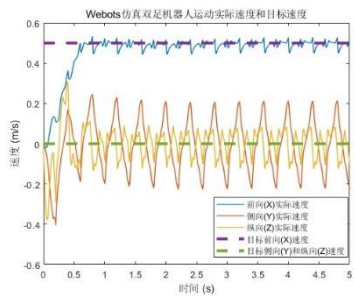
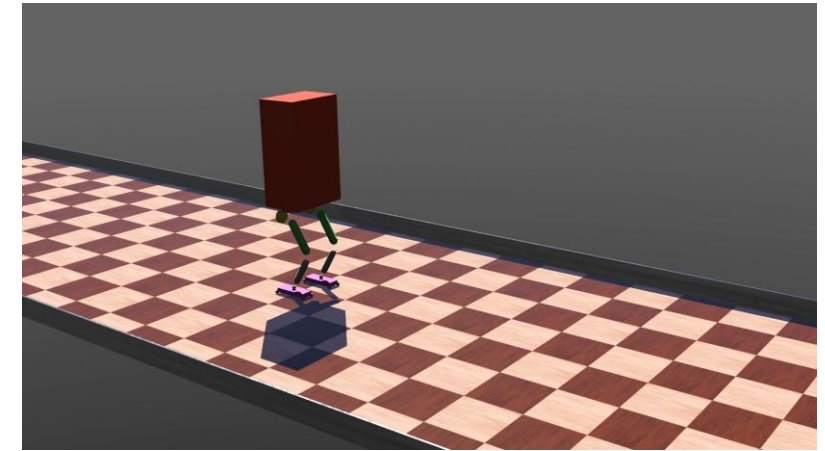
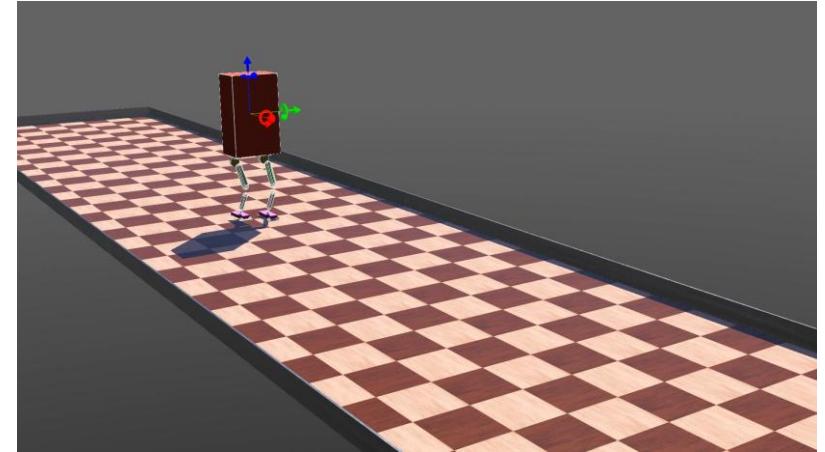
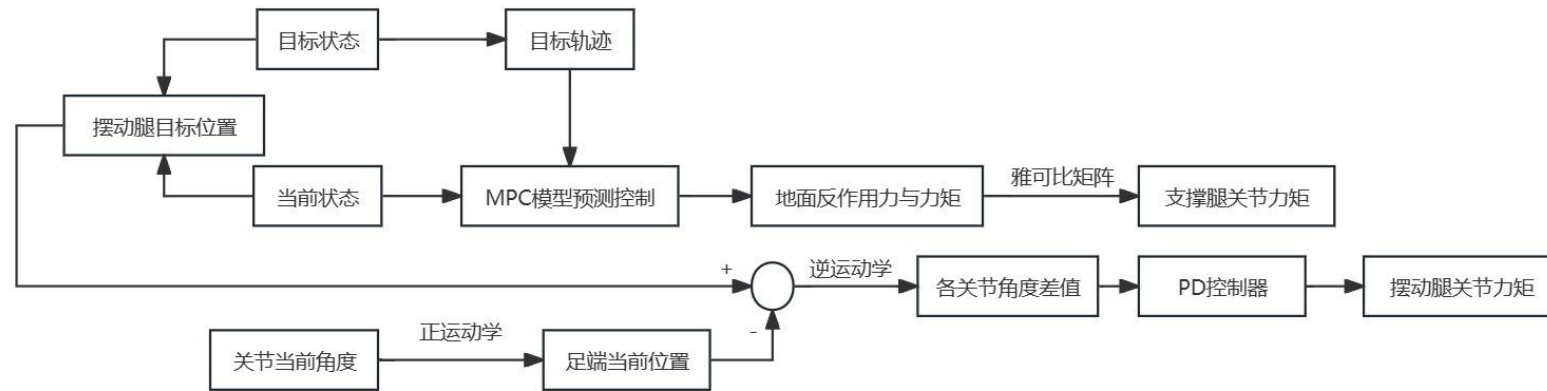


RL Control for Point-Foot Legged Robots

This project develops an RL-based locomotion controller for point-foot legged robots-LimX TRON2. The policy learns to track velocity commands, maintain stable gait patterns, and adapt to slopes, stairs, and uneven terrains. Reward terms include velocity tracking, posture stability, gait regularity, energy efficiency, and contact consistency.



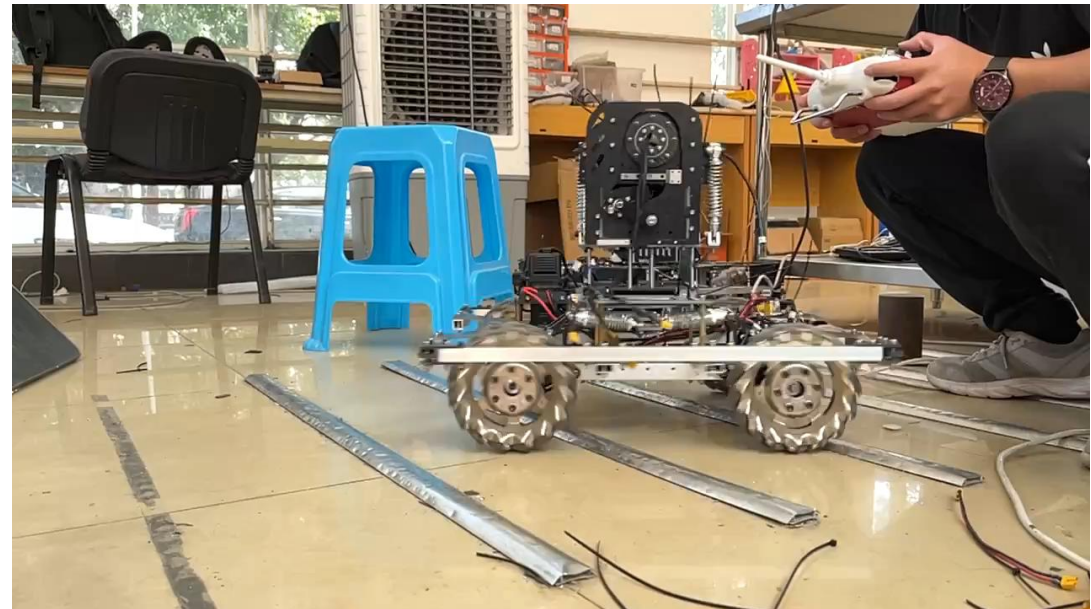
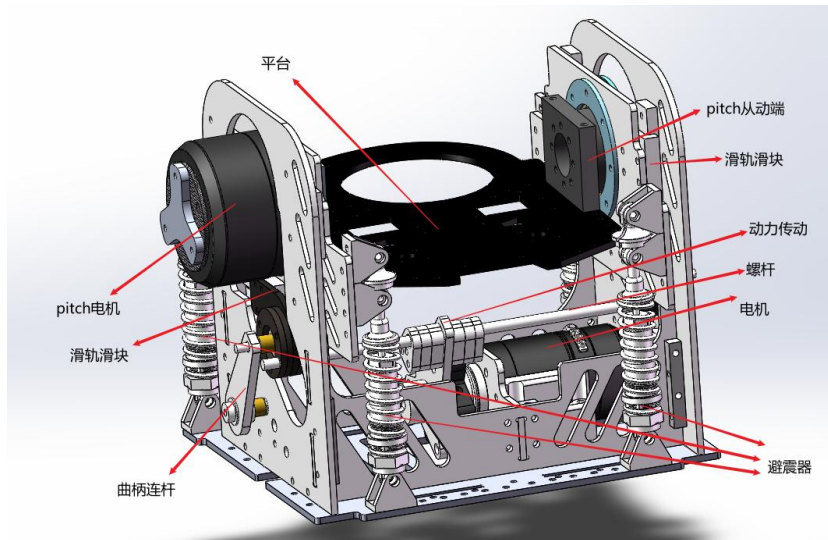
Model Predictive Control System Design and Simulation of Bipedal Robot Gait Control Strategies



Bipedal Robot Gait Control System Block Diagram and Experiment Results with MPC control

Three-Axis Self-Stabilizing Gimbal for Shock Absorption in Fruit and Vegetable Transportation

Developed an STM32 + FreeRTOS based three-axis self-stabilizing gimbal system for vibration reduction and attitude stabilization. Implemented IMU signal acquisition, digital filtering, Kalman-filter-based attitude estimation, and cascaded PID control for real-time disturbance compensation and motor actuation. This project provided hands-on experience in embedded control, signal processing, sensor fusion, and real-time system debugging.





Three-Axis Self-Stabilizing Gimbal for Shock Absorption in Fruit and Vegetable Transportation

24 南方论坛 South Forum 2023年11月下

一种用于农业巡检机器人的云台系统设计与样机实验

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(华南农业大学工程学院, 广东广州 510642)

摘要:【目的】适应复杂多变的农业巡检场景, 减少人工成本和使用成本。【方法】本研究通过综合考虑当前农业巡检领域的发展趋势以及应用需求, 提出了一种满足农业巡检环境高精度、快速响应和轻量化要求的云台系统设计方案。运用串级PID控制算法, 遵循轻量化设计原则, 实现云台系统的性能优化, 有助于确保云台系统在复杂农业环境下的稳定性和灵活性。【结果】通过设计样机进行实验验证, 发现设计的云台系统在控制精度和响应速度方面取得了较好的效果; 该系统能够以毫秒级的响应速度稳定地跟随角度变化, 并同时实现了无超调和无静差的控制效果。【结论】该云台系统的适应性较强, 能够为农业领域的巡检任务带来更高效率和可靠性的技术支持。未来可以进一步完善云台系统的控制策略, 以适应更为复杂多变的农业巡检环境, 进一步推动农业智能化的发展进程, 为实现农业生产的高效、精准和可持续发展作出贡献。

关键词: 农业巡检; PID算法; 双轴云台; 控制系统设计
中图分类号: TP242 **文献标志码:** A

DOI: 10.3969/j.issn.1672-3872.2023.22.007

0 引言

随着巡检机器人的不断研究、发展, 云台的设计已成为当前研究的热门课题之一, 在此发展前景下, 云台的设计广泛应用于巡检机器人、航拍无人机、农业、医疗、监测控制等众多领域。

国内云台已有30多年的发展历史, 最早的云台主要用于摄像机上。经过不断的研究发展, 云台的性能也从最初的单方向控制到现在的全方位360°控制。在农业领域, 大疆T50农业无人机内置高精度、高稳定性的机械式云台, 大大提高了机器的灵活性, 不仅能够进行精确旋转和平移, 还能够实现摄像机的稳定运动, 进行数据采集和控制^[1]。而在国外的云台研究上, 从最开始的只能实现单一方向旋转以及转动精度和承载能力都比较低的状态, 发展到了如今具有一定高度的云台技术。美国菲力尔公司(Teledyne FLIR)研发的FLIR PTU-E46高精度扫描云台仅仅采用了一对步进电机, 由12 V~30 V直流电源驱动, 其俯仰角度为111°, 方位角范围为-180°~+180°, 非常小巧且质量轻便^[2]。目前部分云台系统应用PID算法来对云台进行稳定设计^[3-6]。在确定了相关参数的PID控制下, 可以保证一定工况下的理想效果, 在控制精度以及响应速度上能够满足运行要求^[3]。

基于此, 本文设计了一种控制精度较高、响应速度快、轻量化、适应复杂农业巡检环境的云台系统, 同时设计了农业巡检机器人样机, 对云台系统的控制精度和响应速度进行了实验, 经过实验, 该云台系统

作者简介: 李煜东(2001—), 男, 广东汕尾人, 本科, 研究方向为农业机械及其自动化。

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2023年12月上 South Forum 南方论坛 31

一种丘陵山地农田自适应机械底盘结构设计及试验

黄泽凯, 李煜东, 黄增航, 黄泽灏

(华南农业大学工程学院, 广东广州 510642)

摘要:【目的】适应丘陵山地地形复杂、土地破碎、土壤松软且黏性大的农田行驶场景, 克服这种地形行驶的低稳定性以及行驶通过性较差等问题, 设计一种适用于山地丘陵地形的机械自适应调节减震底盘样机。【方法】研究小组通过采用自适应连杆机构原理, 设计了一种适用于山地丘陵地形的机械自适应调节减震底盘样机, 包括自适应连杆机构、悬挂减震机构等, 对底盘的连杆执行机构和悬挂减震机构参数进行了理论分析计算, 通过对样机模拟的飞坡测试、颠簸路面测试及抗摔刚度测试, 验证了自适应底盘的机械理论分析和计算结果。【结果】1) 飞坡测试: 当整车在空中最高点时底盘与地面夹角为8°~10°, 落地时底盘与地面的夹角为20°~27°时, 此时飞坡姿态最好。2) 颠簸路面测试: 减震带密集程度在0.08 m~0.2 m, 高度差在0.03 m, 整车具有很好的减震效果, 且能够正常作业。3) 抗摔刚度测试: 整车在承受上百次自由落体后未有明显的形变, 仍有较好的刚度及疲劳强度。【结论】自适应调节悬挂可以在不约束车辆自由度的同时, 使得每个车轮间可以根据其他车轮的抬升或下降而做出相应改变, 实现机械上的自适应效果, 大大提高了底盘的行驶通过性以及对接触面的拉力, 对丘陵山地等复杂地形的适应性较强, 能够为未来的丘陵山地的农业领域带来更稳定性和可靠性的机械结构设计支持。

关键词: 农田; 自适应连杆; 减震; 悬挂系统
中图分类号: S220.3 **文献标志码:** A

DOI: 10.3969/j.issn.1672-3872.2023.23.010

0 引言

2021年, 我国农作物耕种收综合机械化率达72%^[1], 丘陵山区农作物耕种收综合机械化率不到50%, 西南丘陵山区仅为29%, 严重制约了全程全面农业机械化^[2]。目前的自走式农业机械底盘在丘陵山区的各种复杂地形的行驶环境下进行探行作业时, 由于道路狭窄、土地破碎, 导致通过性差; 同时丘陵山区土质松软且黏性、湿度大, 南方降水密集, 易形成泥泞的地形, 农机易陷入其中难以脱离^[3-6]。这在一定程度上制约了农业机械化进程, 也阻碍了农业机械化的发展, 这对于当前一般的农业机械来说, 应用和适应性有一定的困难^[3-6]。

近年来, 国内外农业机械底盘研究及设计逐渐呈现“高效率”“稳定性”“智能化”等发展趋势, 在机械底盘的结构设计、作业性能等诸多方面取得了创新性的成果。我国相关研究发展迅速, 现有技术能够适应绝大多数的实际运行环境。张伟军等研发了一款全地形适应性田间作物巡检机器人, 采用主被动复合柔性驱动控制策略, 保障了行进过程中机载激光传感器及摄像头图像获取的稳定性^[7]。Kayacan等为解决轮式底盘在复杂地形中行驶导致其结构适应性明显降低的问题, 研究了智能化自适应底盘, 在自主作业过程中利用机器视觉算法, 令其自动驾驶快速穿越农田地来收集作物

作者简介: 黄泽凯(2001—), 男, 广东汕头人, 本科, 研究方向为机械设计及制造及其自动化。

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物数据, 具有优秀的田间通过能力^[8]。

鉴于此, 本研究提出了一种新型自适应的轮式悬挂减震, 使其设计结构能够很大程度上适应复杂地形的行走通过性以及对接触面的抓地性。

1 底盘结构设计及其原理

1.1 底盘构成

农用自适应机械底盘主要由轮系、摇臂悬挂、底盘框架、减震机构和自适应连杆机构等组成, 底盘结构示意图如图1所示。该底盘框架搭建为“井”字形结构, 同时结合碳纤维板进行连接, 有利于承重和结构的稳定^[9-10]。采用非独立悬挂方式, 悬挂部分采用碳纤维板充当摇臂, 与纵向减震簧通过铰链连接, 增加前后轮系的行程, 采用两两为一组的减震簧簧; 在两侧的簧簧中间用铰链连接一对三角形自适应传动板, 左右两对三角形自适应传动板由一根连杆进行连接, 构成一套自适应连杆执行机构, 整套自适应连杆执行机构固定在铝-碳纤维框架上, 从而实现一侧轮系行程的变化能使另一侧轮系实现自适应调节, 即一侧轮系抬升时另一侧轮系会自适应降低, 以此增加抓地力。该底盘采用四驱供电, 可以根据不同需求配备不同功率的电机以及其他辅助传动部件。

1.2 自适应连杆机构

本研究设计的农用自适应底盘通过连杆机构将左



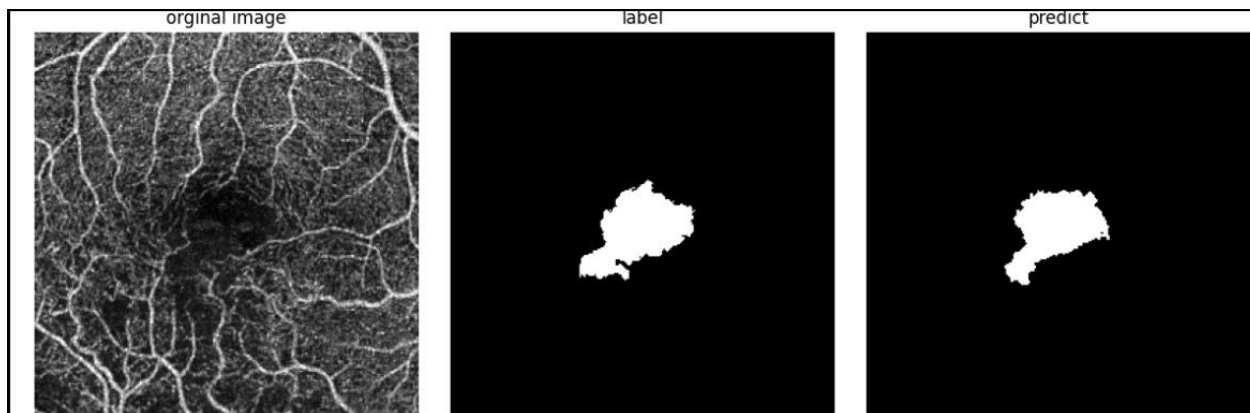
《一种用于农业巡检机器人的云台系统设计与样机实验》第一作者

《一种丘陵山地农田自适应机械底盘结构设计及试验》第二作者

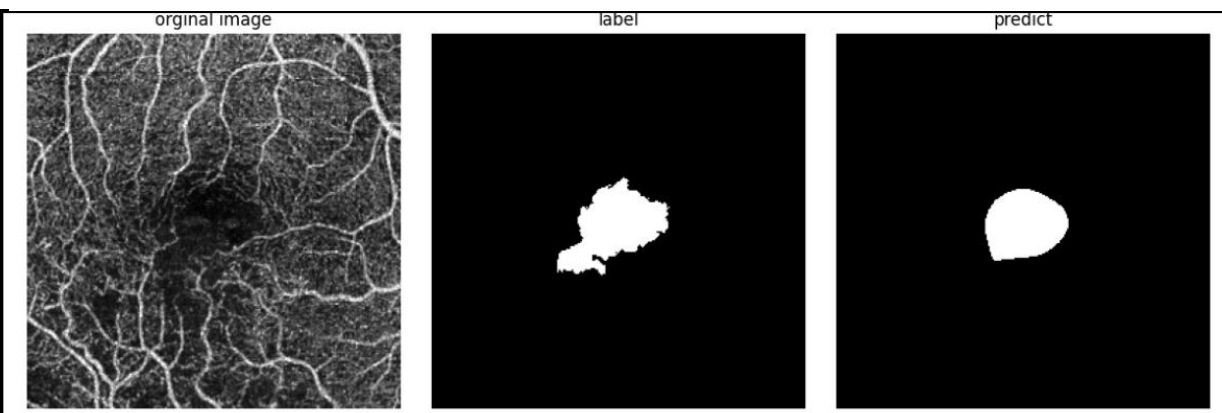
《一种用于果蔬运输的运载缓震自稳平台》实用新型专利

RL-Based Hyperparameter Optimization for TransUNet in 2D Medical Image Segmentation

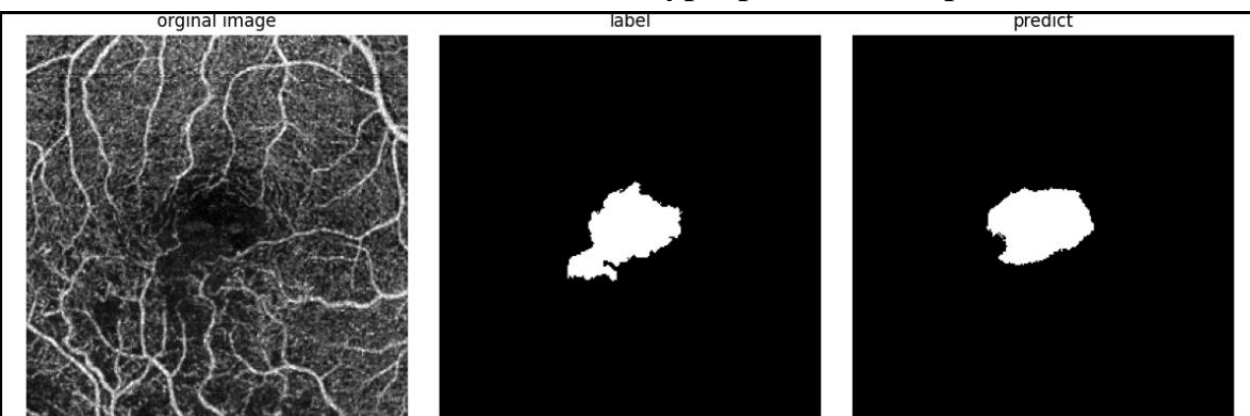
RL is used to adaptively tune TransUNet/UNet hyperparameters, improving training stability and 2D medical image segmentation quality.



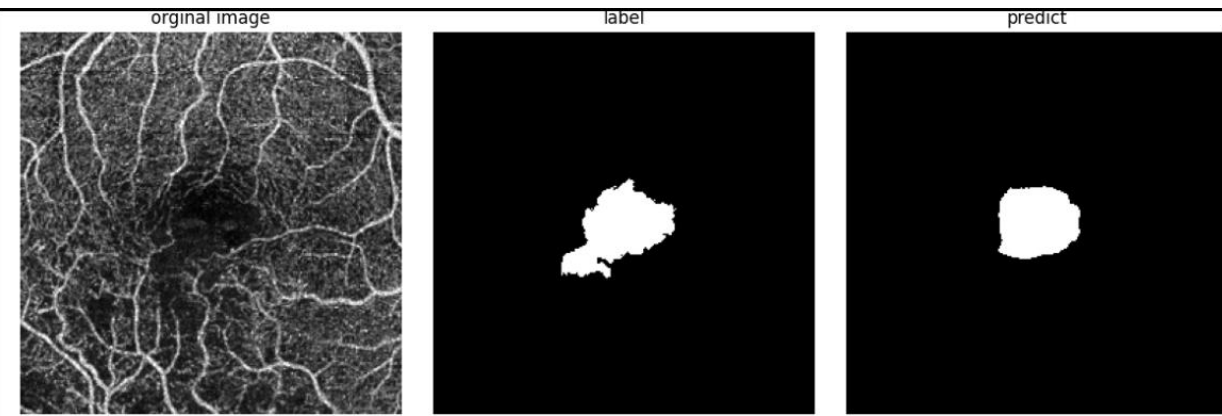
TransUNet with RL-Based Hyperparameter Optimization



TransUNet



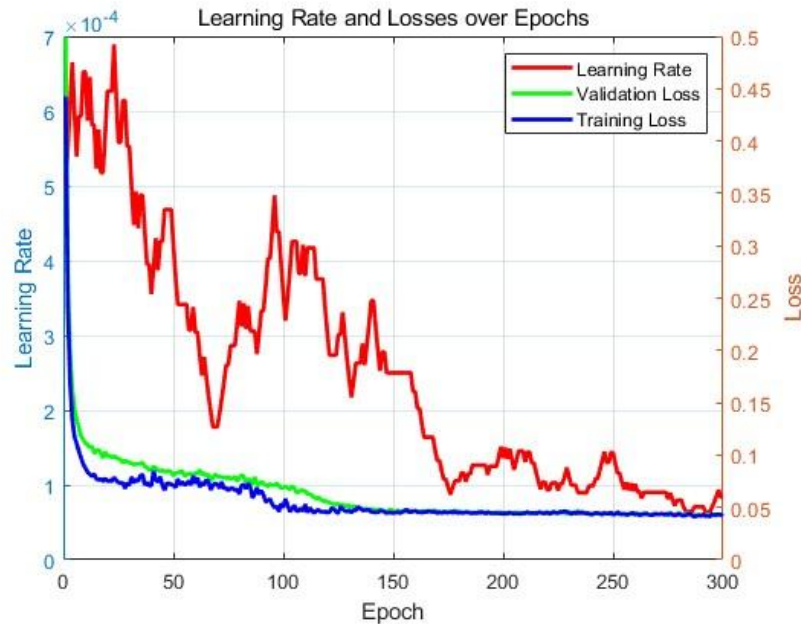
UNet with RL-Based Hyperparameter Optimization



UNet

RL-Based Hyperparameter Optimization for TransUNet in 2D Medical Image Segmentation

RL is used to adaptively tune TransUNet/UNet hyperparameters, improving training stability and 2D medical image segmentation quality.



Learning rate and loss graph with RL-Based Hyperparameter Optimization

	Dice	HSSD	HD95
TranUNet(RL)	0.891	0.5972	3.0000
TrasUNet	0.806	1.9059	8.3066
UNet(RL)	0.789	1.4545	6.4807
UNet	0.709	4.5560	17.088

Domain Adaptation and Generalization in Image Segmentation Based on 2D Fourier Transform

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Abstract—This paper presents a domain adaptation method for medical image segmentation based on the 2D Fourier transform. The approach aims to address the domain gap in segmentation tasks by aligning the frequency components of medical images from various domains, enabling the model to generalize better across different datasets. We applied this method to the TransUNet [1] model and evaluated its performance on the PAZ dataset, which includes images from five distinct domains. The results show that domain adaptation using the 2D Fourier transform significantly improves segmentation accuracy, with the adapted model achieving a higher Dice coefficient and reduced performance variance across domains compared to the baseline model. Our findings suggest that this method provides a promising solution to enhance the robustness and generalization capacity of deep learning models in the segmentation of medical images. However, challenges remain in optimizing the frequency component manipulation parameters and more validation across diverse imaging modalities is needed to assess the broader applicability.

Index Terms—Domain Adaptation, 2D Fourier Transform, TransUNet, Medical Image Segmentation, Generalization

I. INTRODUCTION

A. Background and Motivation

Segmentation of 2D images is a fundamental task in medical image analysis, critical for identifying and differentiating structures and tissues. It plays a vital role in disease diagnosis, treatment planning, and therapy assessment. Deep learning models, such as TransUNet, have demonstrated great promise in segmentation tasks; however, their performance is highly sensitive to domain variations. Differences in imaging modalities, patient demographics, and acquisition conditions often lead to significant accuracy degradation when models are applied to unseen datasets.

To address these challenges, domain adaptation and generalization techniques are essential. The 2D Fourier transform provides an effective approach for manipulating image frequency components, facilitating domain adaptation. By adjusting the magnitude and phase spectra, style and content information can be transferred between domains, reducing the domain gap and improving model generalization. This is especially relevant in medical imaging, where datasets from diverse sources are common.

In this study, we explore the application of 2D Fourier transform-based domain adaptation to enhance the robustness of segmentation models. By leveraging frequency domain manipulation, we aim to improve the performance of models like

TransUNet, ensuring they remain effective even when faced with domain shifts. This work addresses domain discrepancies and lays the foundation for more reliable medical image segmentation in clinical practice.

B. Problem Definition

Despite significant progress in deep learning for medical image segmentation, domain discrepancies continue to challenge model performance. Variations in imaging modalities, patient populations, and acquisition conditions create a gap between training and target domains, leading to performance degradation. While models like TransUNet perform well within their training domains, they often struggle to generalize to unseen datasets.

Traditional methods such as data augmentation and simple domain adaptation techniques frequently fall short in bridging these gaps. They fail to effectively capture distribution shifts, resulting in insufficient adjustments for robust performance across diverse domains. As a result, advanced domain adaptation and generalization techniques are necessary to close the domain gap and improve model performance.

This research focuses on addressing domain adaptation in medical image segmentation by leveraging the 2D Fourier transform. By manipulating image frequency components, we aim to align domain distributions and enhance the generalization capabilities of segmentation models. This approach offers a promising solution to domain discrepancies, enabling models to better adapt to new datasets and maintain robust performance.

C. Contributions

This work presents a novel approach utilizing the 2D Fourier transform for domain adaptation and generalization in medical image segmentation. The key contributions are as follows:

- **Domain Gap Reduction:** We propose a 2D Fourier transform-based method to reduce domain gaps by manipulating the frequency components of images, enabling effective transfer of style and content between domains.
- **Comprehensive Evaluation:** We evaluate the proposed method on 2D medical image segmentation tasks, demonstrating notable improvements in segmentation accuracy across multiple domains. Metrics such as Dice coefficient, ASSD, and HD95 are used to validate the effectiveness of our approach.

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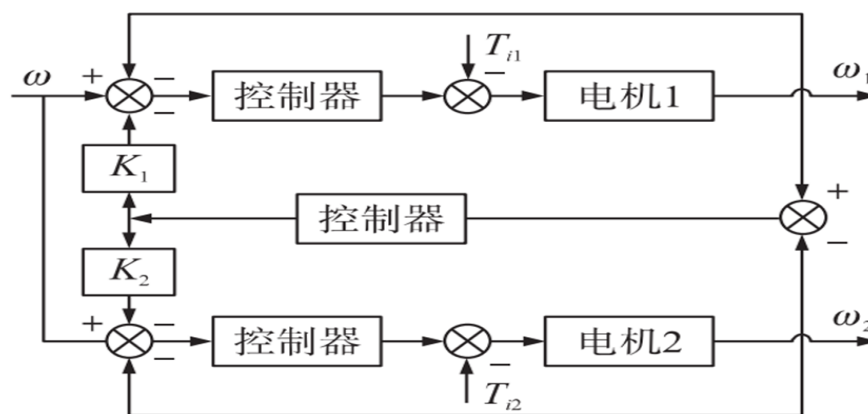
PART FOUR

Competition Experience



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全国大学生机器人大赛 ROBOMASTER 机甲大师系列比赛



- **Competition Awards :**
- ROBOMASTER 2022 机甲大师 超级对抗赛 **全国赛 一等奖**
- ROBOMASTER 2022 机甲大师 超级对抗赛 哨兵机器人组实战奖 **全国 一等奖**
- ROBOMASTER 2022 机甲大师 超级对抗赛 **区域赛**（南部赛区） **一等奖**
- 第八届国际大学生智能农业装备创新大赛 **全国 一等奖**
- 2022年广东省工科大学生实验综合技能竞赛 **省级 一等奖**
- 第二十五届中国机器人及人工智能大赛广东省赛区 **省级 三等奖**



THANKS!

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